Topics in OLS Estimation: Variable selection and misspecification

Rae Tiffen

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Recap of estimation from bivariate regression

- We seek to model the random variable y as a linear function of a constant, random variable x, and random disturbance u. We observe n pairs of (y_i, x_i)
  - $y_i = \beta_0 + \beta_1 x_i + u_i$
- The estimator of β_1 which minimizes the sum of squared residuals (total magnitude of u) is $\hat{\beta}_1 = \frac{\text{cov}(x,y)}{\text{var}(x)}$
- $\frac{dy}{dx} = \hat{\beta}_1$
- $\hat{\beta}$ is the the expected difference in y for a unit larger x
  - (Whether β_1 can be interpreted as the *change* in y for a unit *change* in x is model- and assumption-dependent)
- We are often interested in separating the *partial* effects of *multiple* variables on y.
- What does this mean, and how do we do it?

A quick note on terminology

- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$
  "We regress y on x_1 and x_2"
- A linear model with parameters estimated by OLS. *Not* a "OLS model."
  - Other estimation methods exist (though not in this course)

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Motivation

> Lamb says somewhere that if, of three friends (A, B, and C), A should die, then B loses not only A but "A's part in C," while C loses not only A but "A's part in B." In each of my friends there is something that only some other friend can fully bring out.

– *The Four Loves*, C.S. Lewis

- The approach of OLS is to estimate what part of A is brought out by B and C, what part of B brought out by A and C, and so on
- What "part" of each variable is explained by the others?
- What "part" of education is explained by parental wealth?

Partial effects

- A, B, C as variables in some model of an outcome Y
- We want the partial effect of A
- OLS is concerned with what part of A is attributable to B and C
- Why?
- In finding A's partial effect on the outcome, we are only interested in what *isn't* attributable to B and C
- We want to *control* for B and C

The meaning of controls

- How do we estimate what part of A is attributable to B and C?
- OLS
- Consider an outcome of wages; A, years of education; B, parental wealth; and C, SAT score
- What is the effect of education on wages, controlling for parental wealth?
    1. Regress education on parental wealth, save the residuals
    2. Regress wages on the residuals
- What is the effect of education on wages, controlling for both parental wealth and SAT score?
    1. Regress education on parental wealth and SAT score, save the residuals
    2. Regress wages on the residuals

Frisch-Waugh-Lovell theorem

- Let $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_j x_j + \cdots + \beta_k x_k + u$
- The OLS estimate of any β_j can be found with the two-step process:
    1. Regress x_j on the set of other covariates $1, \ldots, x_k$ and save the residuals $\tilde{x}_j$
    2. Regress y on $\tilde{x}_j$
- $\frac{\partial y}{\partial x_j} = \hat{\beta}_j = \frac{\mathrm{cov}(\tilde{x}_j, y)}{\mathrm{var}(\tilde{x}_j)}$
- In words: the coefficient $\hat{\beta}_j$ reflects the partial effect of x_j on y by removing from x_j the part explained by the other covariates.
- This was first shown in 1907, and in econometrics is attributed to Frisch & Waugh's 1933 paper

FWL example

- In the CARD dataset:
  - `reg wage educ`
    $\text{wage} = 183.9 + 29.7(\text{educ})$
  - `reg wage educ age fatheduc`
    $\text{wage} = -566 + 20.8(\text{educ}) + 28.3(\text{age}) + 8.2(\text{fatheduc})$
- What is the effect of controlling for age and father's education on the coefficient on education?
  1. `reg educ age fatheduc`
     $\text{educ} = 8.7 + 0.5(\text{age}) + 0.336(\text{fatheduc})$
     `predict educ_tilde, residuals`
  2. `reg wage educ_tilde`
     $\text{wage} = 589 + 20.8(\text{educ})$

Double residual regression and the partial R^2

- $\frac{\text{cov}(\tilde{x}_j, y)}{\text{var}(\tilde{x}_j)} = \frac{\text{cov}(\tilde{x}_j, \tilde{y})}{\text{var}(\tilde{x}_j)}$
- In the double residual case, we also remove from y the part explained by the other (not x_j) covariates
- $\hat{\beta}_j = \text{corr}(\tilde{x}_j, \tilde{y}) \times \frac{\sigma_{\tilde{y}}}{\sigma_{\tilde{x}_j}}$
- In this case:
    1. The residuals are the same as in the full regression of y on $1, \ldots, x_j, \ldots, x_k$
    2. The R^2 of the double residual regression is the partial R^2 of x_j in the full regression (in Stata, you can find this with `pcorr`)
    3. The homoskedasticity-assuming standard errors of $\hat{\beta}_j$ differ from the full regression by a degrees of freedom adjustment

Frisch-Waugh-Lovell Theorem for standard errors

- Recall the homoskedasticity-assuming standard error in the bivariate case $(y = \beta_0 + \beta_1 x_1 + e)$
  - $\mathrm{se}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\mathsf{SST}_x}} = \sqrt{\frac{1}{n-1}\frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$
- In the regression of y on $1, \ldots, x_j, \ldots, x_k$, the standard error of any $\hat{\beta}_j$ is:
  - $\mathrm{se}(\hat{\beta}_j) = \sqrt{\frac{\hat{\sigma}^2}{\mathsf{SST}_{\tilde{x}_j}}} = \sqrt{\frac{1}{n-k-1}\frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n \tilde{x}_i^2}}$
- This can be rewritten as:
  - $\mathrm{se}(\hat{\beta}_j) = \sqrt{\frac{1}{n-k-1}\frac{RSS}{TSS_{x_j}(1-R_j^2)}}$
- Where:
  - TSS_{x_j} is the total sum of squares of x_j (NOT $\tilde{x}_j$), and
  - R_j^2 is the R^2 from the first-step regression of x_j on the other covariates
- What does the $(1 - R_j^2)$ in the denominator tell us?

FWL example continued

- `reg wage educ age fatheduc`
  $\text{wage} = -566 + \underset{(2.16)}{20.8}(\text{educ}) + 28.3(\text{age}) + 8.2(\text{fatheduc})$
- Generate $\widetilde{\text{wage}}$ and $\widetilde{\text{educ}}$ by regressing each on age and fatheduc, then saving the residuals as new variables
- `reg wage_tilde educ_tilde`
  $\widetilde{\text{wage}} = \underset{(2.16)}{20.8}(\widetilde{\text{educ}})$
- The SEs differ by a degree of freedom adjustment, which in the above case where $n = 2,320$ is negligible.
- The R^2 in the full regression is 0.1554, and in the double residual regression is 0.0386. How do we interpret this?

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Multicollinearity

- Recall the quote about 3 friends, A, B, and C
- If B is always with C — they're practically glued to each other — it's difficult to know how B would react to something independent of C
- When one variable is almost entirely explained by the other covariates, very little variation remains after we control for them
- Example: SAT scores and ACT scores. Both seek to measure latent ability
  - How do we interpret the residuals from the regression of SAT score on ACT score?
  - A hint: If they are both *perfect* measures of ability, what will the R_j^2 for SAT scores be?

Variance "inflation" factor

- Recall the estimator of variance
  $\text{var}(\hat{\beta}_j) = \frac{1}{n-k-1} \frac{RSS}{TSS_{x_j}(1-R_j^2)}$
- If a variable is almost entirely explained by the other covariates, the R_j^2 will be very large
- In this case, how does the variance in the full regression (y on all covariates) compare to the simple regression of y on x_j alone?
- It "inflates" by a factor of $\frac{1}{1-R_j^2}$

Understanding standard errors

- $\text{se}(\hat{\beta}_j) = \sqrt{\frac{1}{n-k-1} \frac{RSS}{TSS_{x_j}(1-R_j^2)}}$
- What are standard errors a function of?
    - *Sample size minus estimated parameters:* larger sample size $\rightarrow$ smaller standard error
    - *Residual sum of squares:* smaller residual variance $\rightarrow$ smaller standard error
    - *TSS*$_{x_j}$*:* greater variance in x_j $\rightarrow$ smaller standard error
    - *VIF:* lower R_j^2 (prop. of var. explained by other covariates) $\rightarrow$ smaller standard error
- SEs refer to the distribution of the estimator
- Interpretation-dependent: removing a covariate may lower the R_j^2, but it changes the coefficient's interpretation (and may also change RSS)

Multicollinearity in practice

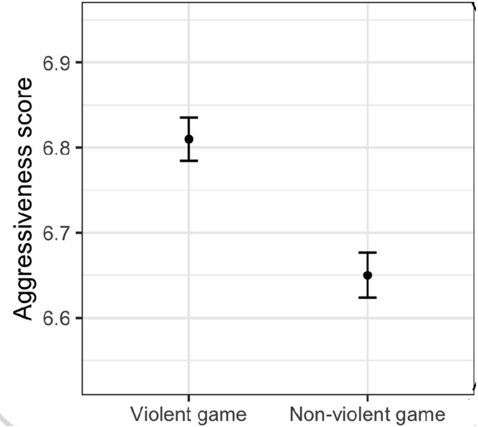
- Multicollinearity is not a statistical problem, it is a fact of the world. Related things will be correlated
- The more variables you control for, the more variation you remove, and the less you have to estimate β_j
- This is no more or less serious than small sample sizes. Both with small samples and high collinearity, you have less information to estimate your parameters
- Arthur Goldberger (1930–2009): "To say that 'standard errors are inflated by multicollinearity' is to suggest that they are artifically, or spuriously, large. But in fact they are appropriately large: the coefficient estimates actually would vary a lot from sample to sample. This may be regrettable but it is not spurious."
- What are some examples? How would you interpret the remaining variation?

A caution about standard errors

- In our earlier example, the coefficient on education was 20.8 with a standard error of 2.16. How do you interpret this?

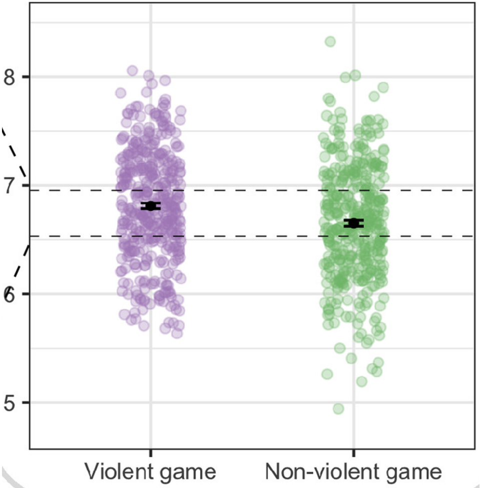
A caution about standard errors

One SE above/below the estimated mean:



What is the probability that, if you picked one violent game and one non-violent game at random, the violent game would have a higher aggressiveness score?

A caution about standard errors



58%. More data $\rightarrow$ more certain estimate, doesn't alone tell us about the outcome distribution.

(figures from Zhang et al., "An illusion of predictability in scientific results", 2023)

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Recap of FWL — multiple regression estimation

- Let $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_j x_j + \cdots + \beta_k x_k + u$
- Regress x_j on the set of other covariates:
  $x_j = \underbrace{\alpha_0 + \alpha_1 x_1 + \cdots + \alpha_k x_k}_{\text{part of } x_j \text{ explained by other covariates}} + \underbrace{\tilde{x}_j}_{\text{part left over}}$
- Then, regress y on $\tilde{x}_j$ to estimate $\hat{\beta}_j$:
  $\hat{\beta}_j = \frac{\text{cov}(\tilde{x}_j, y)}{\text{var}(\tilde{x}_j)} = \frac{\partial y}{\partial x_j}$
- Main idea: OLS estimates partial effects by removing from each right-hand-side variable the component explained by the other right-hand-side variables
  - And the $\hat{\beta}$ reflects the relationship between y and the part left over
- The interpretaton of each $\hat{\beta}$ is thus heavily dependent on what other variables you include

Recap of multicollinearity — multiple regression standard errors

- The more variables on the right-hand-side, the more variation you "remove" from each variable
- Less variation $\rightarrow$ less precise estimates
- $\text{se}(\hat{\beta}_j) = \sqrt{\frac{1}{n-k-1} \frac{RSS}{TSS_{x_j}(1-R_j^2)}}$
- The variance "inflation" factor is $\frac{1}{1-R_j^2}$, where R_j^2 is the proportion of variance in x_j explained by the other right-hand-side variables
- If you remove right-hand-side variables to get a lower R_j^2, the interpretation changes. The larger $\text{se}(\hat{\beta}_j)$ is not artificial.
- "Perfect multicollinearity": $R_j^2 = 1$, no variation left over, OLS will not return estimates

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OVB with one omitted variable and one included variable

- Long: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$
- Short: $y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{u}$
- What is the difference between β_1 and $\tilde{\beta}_1$?
  - Equivalently: what is the effect of including x_2 in the model?
- $\tilde{\beta}_1 = \frac{\text{cov}(x_1,y)}{\text{var}(x_1)} = \frac{\text{cov}(x_1,\beta_0+\beta_1x_1+\beta_2x_2+u)}{\text{var}(x_1)} = \beta_1 + \beta_2\frac{\text{cov}(x_1,x_2)}{\text{var}(x_1)}$
- $\tilde{\beta}_1 - \beta_1 = \underbrace{\beta_2\frac{\text{cov}(x_1,x_2)}{\text{var}(x_1)}}_{bias}$
-

Bias of $\tilde{\beta}_1$	$\text{cov}(x_1,x_2) > 0$	$\text{cov}(x_1,x_2) < 0$
$\beta_2 > 0$	positive bias	negative bias
$\beta_2 < 0$	negative bias	positive bias

- If either β_2 or $\text{cov}(x_1,x_2)$ is 0, the difference between β_1 and $\tilde{\beta}_1$ is also 0.

OVB in the general case

- Long: $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \cdots + \beta_p x_p + u$
- Short: $y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \cdots + \tilde{\beta}_k x_k + \tilde{u}$
- k included variables, p omitted variables
- $\tilde{\beta}_1 = \beta_1 + \underbrace{\sum_{i=k+1}^{p} \alpha_i \beta_i}_{bias}$
- Where:
  - α_i is the coefficient on x_1 in the regression of the omitted variable x_i on the set of included variables $\{x_1, \ldots, x_k\}$
  - β_i is the coefficient on x_i in the long regression
- Including a new variable changes the bias associated with all other omitted variables (changes the α)
  - Which means adding even good controls does not guarantee a reduction of bias

OVB in the general case continued

- $\tilde{\beta}_1 - \beta_1 = \sum_{i=k+1}^{p} \alpha_i \beta_i$
- This can be simplified a bit: by the FWL, the α come from the regressions of each omitted variable on x_1^*, the residuals from the regression of x_1 on the set of included variables
- Would need to know:
  - What is the relationship between x_1^* and the omitted variable?
  - What is the relationship between y and x_i', the residuals from the regression of the omitted variable x_i on $\{x_1, \ldots, x_k, \ldots, x_p\}$?
- Is this feasible to speculate on? Rarely
- The point of this: it gets complicated *fast*

OVB in practice

De Luca et al., "Balanced Variable Addition in Linear Models" 2018:

> "Ludwig van Beethoven composed nine symphonies. Suppose a 10th symphony is discovered. There is no full score, only three parts are available: first violin, cello, and clarinet. This version is recorded and creates a big hit. Of course everybody realizes that many instruments are missing – still, it seems one gets a good idea of Beethoven's tenth. Now the trumpet part is discovered and a new recording is made. The new recording is received less enthusiastically than the first, and music experts claim that adding the trumpet moves us away from how the real symphony should sound.
>
> This creates a puzzle and a debate among scientists of various disciplines. How is it possible that getting closer to the true instrumentation does not get us closer to the true sound? Of course, adding all instruments to the score creates the true sound, but it seems that adding only some of them may not lead to an improvement. An addition in itself is not necessarily an improvement, it must be a "balanced addition.""

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The Outcome-Treatment-Controls paradigm

- What we'll be doing for the rest of the course
- Let Y be some outcome variable, D a variable indicating treatment, and $\mathbf{X} = \{X_1, \ldots, X_k\}$ a set of control variables
- By the FWL, the regression of Y on D and $\mathbf{X}$ is equivalent to:
  1. Regressing D on $\mathbf{X}$, saving residuals $\tilde{D}$
  2. Regressing Y on $\tilde{D}$
- Step 1 is our model of treatment: we remove the variation in treatment explained by $\mathbf{X}$.
  - Corollary: The more controls in $\mathbf{X}$, the less variation remains in D
- We intend to select $\mathbf{X}$ such that the relationship between Y and $\tilde{D}$ can be interpreted causally
- What can be formulated in this paradigm?
- The main idea: We care about *one* coefficient being unbiased

A brief note on "good" and "bad" controls

- Common causes: variables which cause both Y and D
  - Variation we often want to remove
  - Example: education and wages are both influenced by parental wealth
- Colliders: variables which Y and D both cause
  - From Angrist & Pischke's "Mostly Harmless Econometrics": "Bad controls are themselves outcome variables in the notional experiment at hand. That is, bad controls might just as well be dependent variables too."
- Whether a control is "good" or "bad" is context-dependent
- "Good" controls are not guaranteed to decrease bias
- The main point: be thoughtful about what variation you are removing from D

Where are we?

- So far, we've seen how:
- OLS estimates the partial effect of a variable on the outcome by removing the part of the variable explained by the other covariates
- Our selection of covariates determines the interpretation of each coefficient
- The more variables we control for, the less variation that remains in our treatment variable (and the less precise the estimate)
- Omitted variable bias can become complicated very quickly

Considerations when selecting variables

- Select $\mathbf{X}$ such that the relationship between Y and $\tilde{D}$ can be interpreted causally
- What does this even mean? How can we claim to mimic natural experiments with observational data?
- The second half of the course: quasi-experimental designs
- To keep in mind:
  - By adding a new control, what variation are you removing from D?
  - How do you interpret the remaining variation?
  - What is the difference betwen the data you observe and the ideal dataset to answer your question?
- Causal inference requires subject-area knowledge and good research design

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Sensitivity and the “strength” of an omitted variable

- How “strong” would an omitted variable need to be in order to overturn your results?
- Consider the first-step FWL regression for D, the regression of D on $\mathbf{X}$
  - In a way, our model of treatment
- If you’re missing a confounder that you expect explains a large proportion of both treatment and the outcome, what can you say about the bias?

What your computer is actually doing

- Maaaaaaaany OLS algorithms exist
- Modified Gram-Schmidt: basically iterated Frisch-Waugh-Lovell
- $y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$
- Regress x_1, x_2 each on x_3, saving residuals $\tilde{x}_1, \tilde{x}_2$
- Regress $\tilde{x}_1$ on $\tilde{x}_2$, saving residuals x_1^*
- Regress y on x_1^*, estimating β_1

Geometric interpretation of OLS

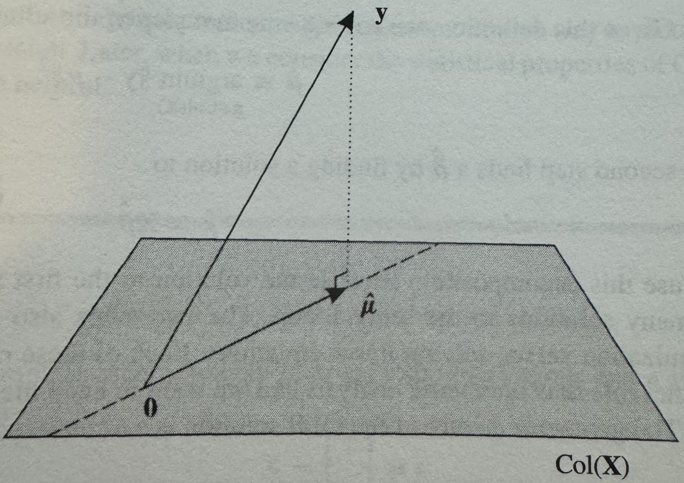

Figure 2.3 Orthogonal projection.